

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456115.7543 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.023
Transit depth (Rp/Rs)^2	2.56 +/- 0.74 [%]
Orbital Inclination (inc)	86.22 +/- 1.17
Airmass coefficient 1 (a1)	0.857 +/- 0.017
Airmass coefficient 2 (a2)	0.131 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.98 %
Best Comparison Star	#3 - [239, 335]
Optimal Method	PSF photometry
Transit Duration (day)	0.0671 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.16 ± 0.023 vs 0.1326 (deviation 1.06σ)
Duration fit vs published	0.0671 d vs 0.0737 d (deviation 1.16σ)
Mid-time O-C	-6.0 min
Depth SNR	6.2
Residual scatter	1.98 %

Light curve

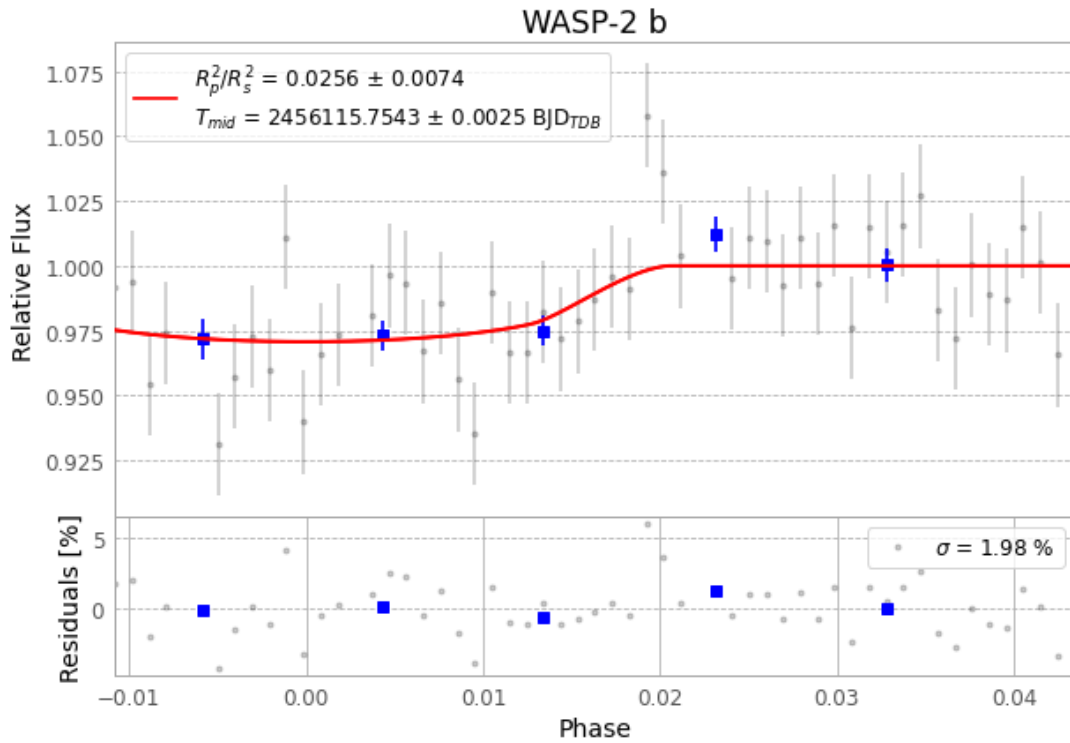


Figure 1 — FinalLightCurve_WASP-2 b_2012-07-06.png (SHA-256 b0733a60939351b0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [196, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5fceb9b591ed68b7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), WASP-2120707052412.FITS → WASP-2120707081213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-120707.log (09e2fe627880...), FinalLightCurve_WASP-2 b_2012-07-06.png (b0733a609393...), FinalLightCurve_WASP-2 b_2012-07-06.csv (e95b648ebf14...)

Compute resources

Wall time 277.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 01:26Z.